GRADUATION PROJECT

When Will Your Food Arrive?

From raw data processing and exploratory analysis to training optimized machine learning models — and constructing an interactive, real-world web application.

PROJECT TEAM

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SUPERVISED BY

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PRESENTER: HABIBA

Presentation Agenda

1. Introduction

Problem context, business importance, and core project targets.

Presented by: Habiba

2. Data & Prep

Dataset overview, missing values imputation, and IQR outlier cleansing.

Presented by: Aliaa

3. EDA Insights

Distribution stats, key correlations, and environmental factor impacts.

Presented by: Youssef Moussa

4. ML Modeling

Feature scaling, pipeline construction, and comparative model training.

Presented by: Mostafa

5. Evaluation & Winner Selection

Rigorous test metric comparisons (MAE, R^2), mathematically proving why the simpler Linear Regression model outperformed complex forests in our dataset.

Presented by: Aliaa

6. Live GUI Web Application

Deploying the trained model live to the browser for interactive, serverless prediction, with real-time analytics graphs.

Presented by: Youssef Shabban

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Problem & Project Goal

The ETA Bottleneck

In modern on-demand logistics, delivery time is the primary determinant of customer loyalty. Inaccurate ETAs lead to food deterioration, rider scheduling overheads, and higher complaints.

Our Objectives

Build a robust Machine Learning engine to predict precise trip durations (minutes). Compare distinct mathematical frameworks, select the highest-performing architecture, and deploy it to a live interface.

Project Deliverables

Cleaned data pipeline from 994 real-world orders. A predictive ML API achieving >85% R^2 validation accuracy, connected directly to a high-fidelity bilingual (AR/EN) web application.

Dataset Overview

FEATURE NAME	TYPE	DESCRIPTION
Distance_km	Numeric	Radial distance to delivery point
Preparation_Time_min	Numeric	Kitchen cooking overhead duration
Traffic_Level	Categoric	Density conditions (Low / Med / High)
Weather	Categoric	Atmospheric state (Clear, Rainy...)
Courier_Experience_yrs	Numeric	Years active of assigned driver

Historical Record Audit

- **Dataset File:** Analyzed raw records inside `Food_Delivery_Times.csv` containing 1,000 distinct delivery logs.
- **Feature Landscape:** 9 raw attributes covering environmental, vehicle, temporal, and human experience metrics.
- **Target Variable:** `Delivery_Time_min` — the continuous variable indicating exact minutes from restaurant to destination doorstep.

Data Cleaning & Prep

1. Handling Missing Data

Categorical attributes with missing entries (Weather, Traffic, Period) were imputed utilizing the statistical **Mode** of the respective groups to maintain consistent distributions. Numerical gaps (Courier Experience) were filled using the robust **Median** metric to prevent skewing.

2. Outlier Removal (IQR Rule)

To eliminate anomaly noise (e.g. system faults, cancelled orders showing extreme delivery durations), we calculated the Interquartile Range (IQR). Rows beyond standard bounds were pruned, purifying the dataset from 1,000 down to **994 clean records**.

3. Categorical Encodings

Converted `Traffic_Level` into ordered ordinal values (0, 1, 2). Applied **One-Hot Encoding** on nominal features (Weather, Time of Day, Vehicle) while dropping the first column to protect models from multi-collinearity issues.

4. Standardized Scaling

Applied `StandardScaler` on numerical features to assure equal weight allocation during training. To prevent critical data leakage, scaling parameters were fit purely on training sets prior to model consumption.

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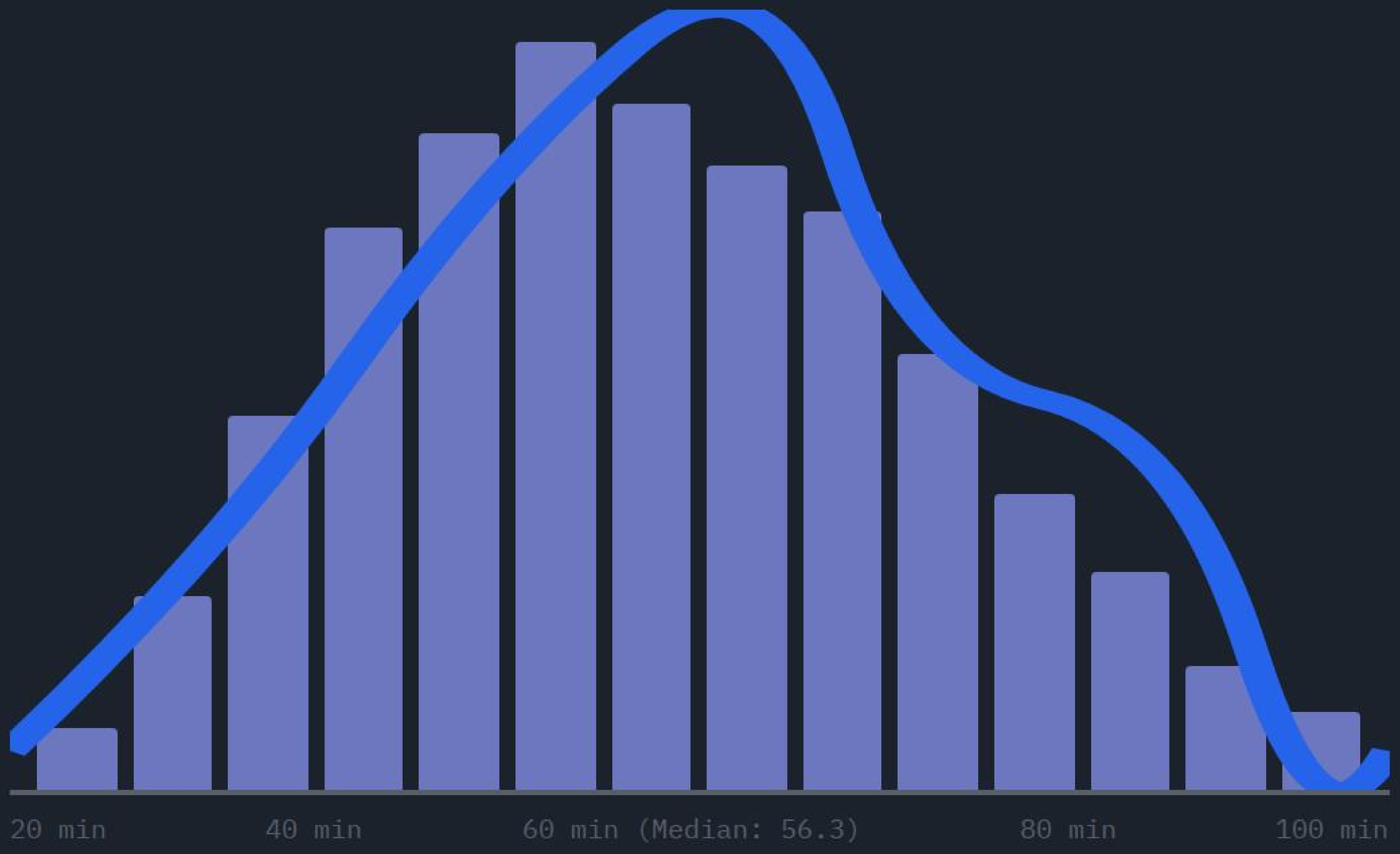
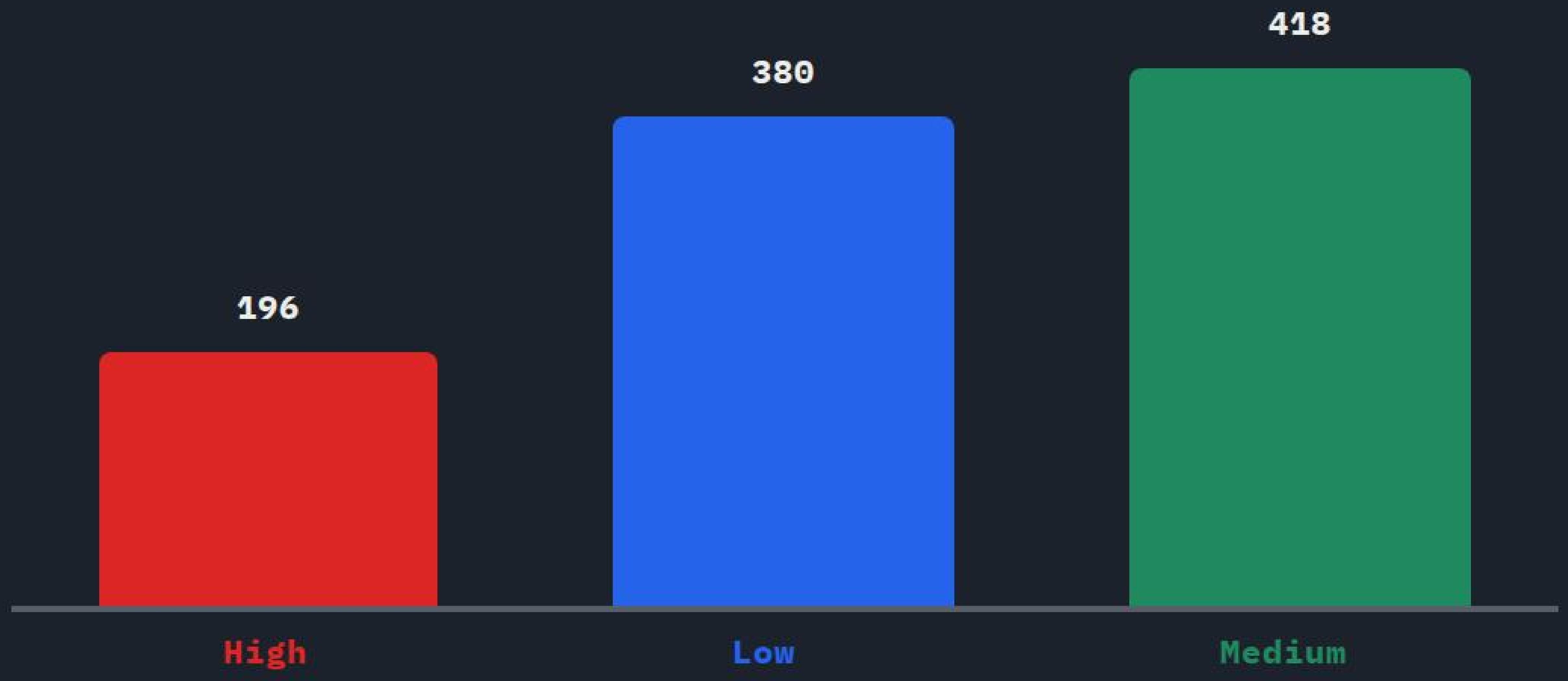
EDA - Distribution Analytics

DELIVERY TIME DISTRIBUTION



Gaussian Distribution: Target variable analysis reveals a symmetrical curve of delivery times, showing optimized mean performance at **56.3 minutes**.

DISTRIBUTION OF TRAFFIC LEVELS



Friction Landscape: High traffic segments constitute only ~20% of occurrences, whereas Medium and Low densities dominate the operational dataset.

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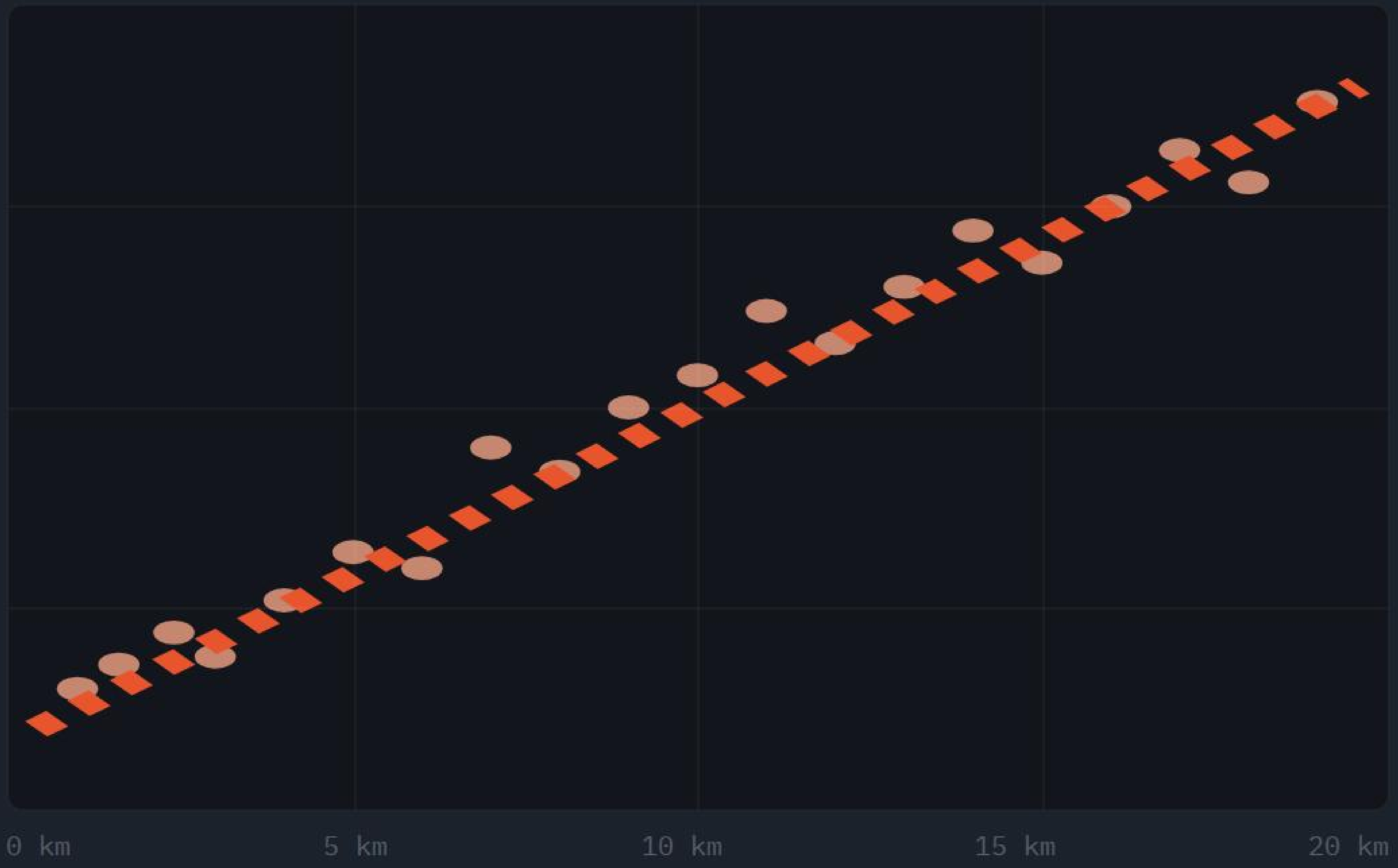
EDA - Distance vs. Delivery Time

RELATIONSHIP: DISTANCE VS DELIVERY TIME



Highly Linear Relationship: Trip distance serves as the absolute leading numerical factor (0.79), scaling in a direct proportional curve with total trip minutes.

CORRELATION MATRIX OF NUMERIC FEATURES

	ID	Dist	Prep	Exp	ETA
Order_ID	1.00	-0.03	-0.03	0.01	-0.04
Dist_km	-0.03	1.00	-0.01	-0.01	0.79
Prep_min	-0.03	-0.01	1.00	-0.03	0.31
Exp_yrs	0.01	-0.01	-0.03	1.00	-0.09
Delivery	-0.04	0.79	0.31	-0.09	1.00

Mathematical Alignment: While distance (0.79) and preparation time (0.31) add sequential overhead, courier experience (-0.09) drives slight inverse duration optimization.

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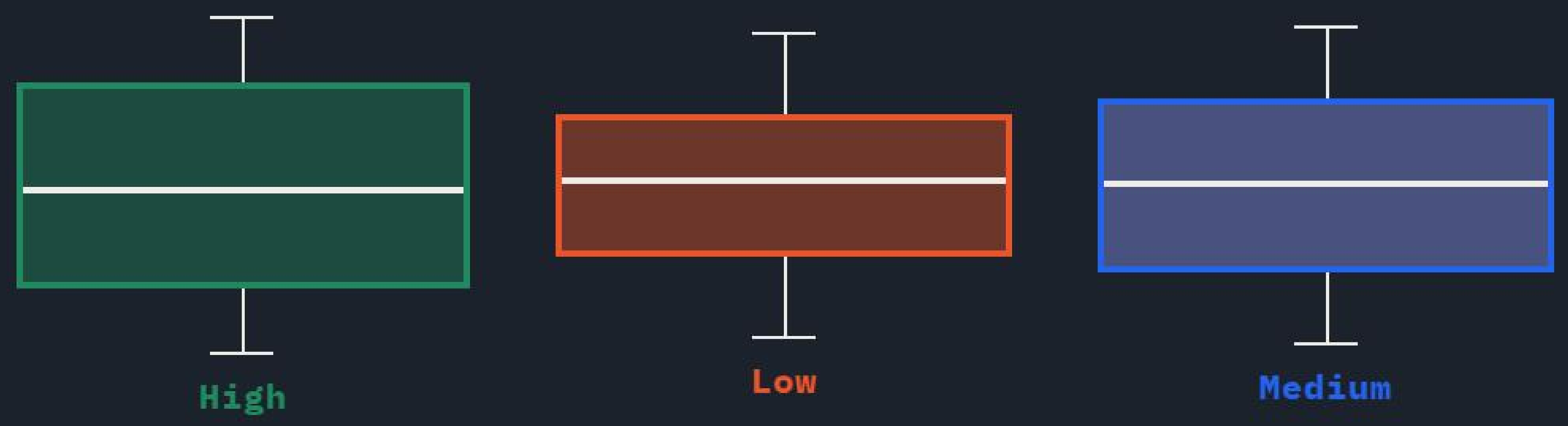
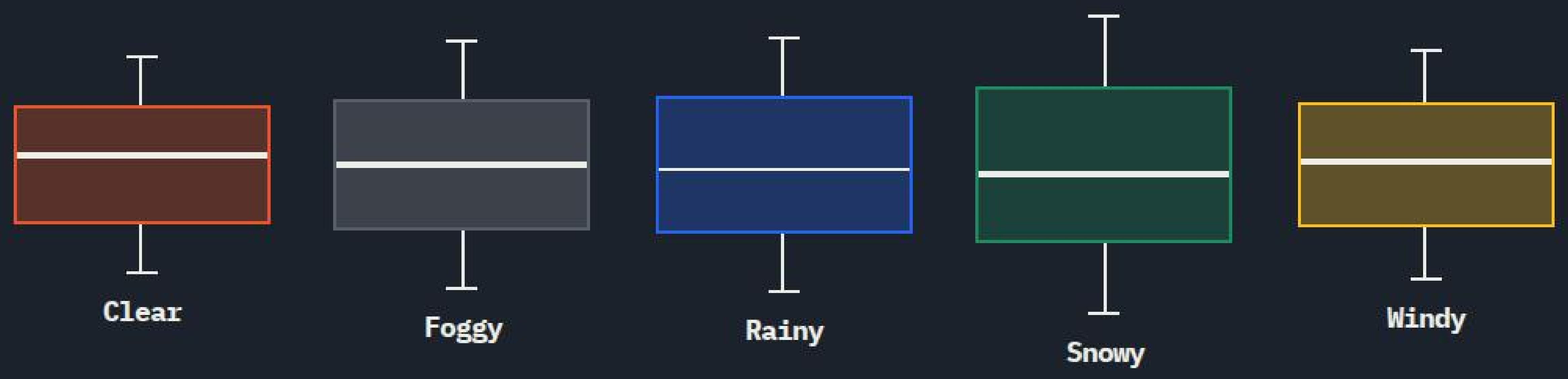
EDA - Environmental Factors

TRAFFIC'S EFFECT ON DELIVERY TIME



Friction Bottlenecks: Transit delay spreads directly by **12–15 minutes** when scaling from Low to High density environments.

WEATHER'S EFFECT ON DELIVERY TIME



Severe Friction: Harsh circumstances (Rainy/Snowy) limit motorcycle maneuvering, causing total duration averages to expand by **~12%**.

Feature Engineering Pipeline

Ordinal Encoding

Traffic level holds a natural sequential weight. We mapped it ordinally to mirror real-world friction progression: `Low = 0`, `Medium = 1`, and `High = 2`.

One-Hot Dummy Vectors

Nominal factors (Weather, Period, Vehicle Type) were split using One-Hot encoding to preserve model mathematical impartiality. We dropped first dummy columns to prevent collinear model bias.

Standard scaling

Adjusted numerical inputs (Distance, Prep Time, Experience) using StandardScaler so that varied numeric spans are evaluated on a balanced standardized variance metric scale.

Data Partition Strategy 80% Train Set (795 Orders) / 20% Holdout Test Set (199 Orders)

Machine Learning Models

Model A: Multiple Linear Regression

Assumes a linear, additive relationship between inputs and delivery times. It models output mathematically by multiplying the learned coefficients with standard input factors. Exceptionally lightweight, runs in milliseconds, and provides absolute parameter interpretability.

```
scikit-learn.linear_model.LinearRegression
```

Model B: Random Forest Regressor

An ensemble bootstrap decision tree regressor that operates by building 100 deep decision trees during training. It outputs the mean predictive time of all individual trees. Excellent at capturing potential non-linear relationships, but demands more compute power.

```
ensemble.RandomForestRegressor(n_estimators=100)
```

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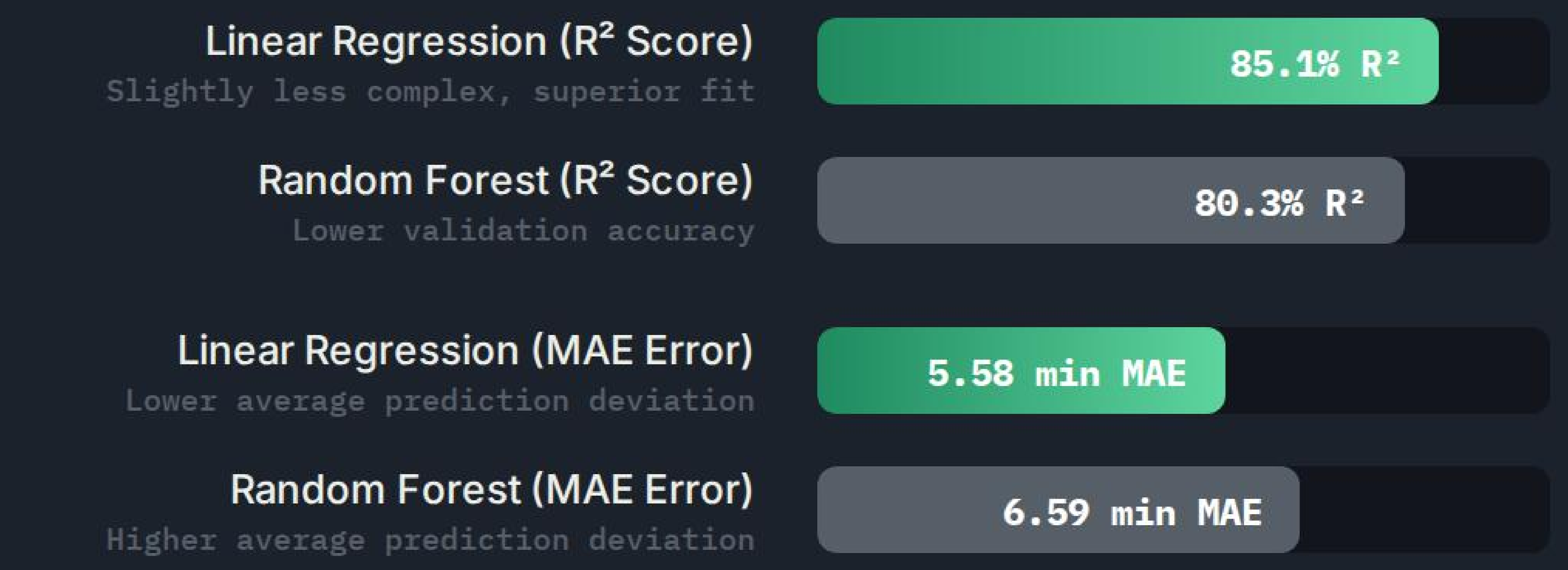
Model Evaluation & Selection



Selection Proof & Reasoning

- **Winner:** Multiple Linear Regression outperformed Random Forest by 4.8% on R^2 and reduced prediction error by a full minute.
- **Occam's Razor:** Since the underlying relationship between distance and delivery time is highly linear, the simpler model fit the data perfectly without overfitting.

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} y_i - \hat{y}_i \qquad R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

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From Notebook to Live Website

- **Real-Time Predictions:** The model computes continuous ETAs live inside client browsers using the trained formula coefficients.
- **Dynamic Sliders & Options:** Users adjust distance, prep duration, traffic congestion, and weather variables to inspect changes in real-time.
- **Full Bilingual Support:** Built-in toggle allows users to switch between Arabic and English instantly.
- **Interactive EDA Dashboard:** Active analytical charts display dataset trends, vehicle counts, and correlation matrix details.

Conclusion & Future Work

Production Recommendations

We recommend deploying the **Multiple Linear Regression** model to production. It maintains a high validation accuracy (85.1%), runs with zero compute latency, and provides complete feature coefficients explaining exact delivery influences to operational stakeholders.

Next Development Iterations

We propose integrating **live mapping APIs** (e.g. Google Maps API) to fetch actual GPS routes. Furthermore, real-time traffic data feeds should replace categorical levels. Finally, setting up continuous training pipelines will combat model drift over time.

PRESENTER: ALL

Thank You!

We welcome any questions or technical inquiries regarding the project.

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